

# Deictic Gesture Grounding: A Six-Condition Ablation Using Frontier Vision Language Models on a Quadruped Robot

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## Abstract

People routinely refer to objects with underspecified language and a pointing gesture: "pick that up" is recoverable only by combining the words, the arm's aim, and what's visible in the scene. This report presents a deictic grounding system on a Boston Dynamics Spot and a controlled comparison of that system against three frontier vision-language models on an identical frame under three prompt conditions. The robot pipeline runs fully offline, recovers the pointing ray from Spot's own camera with YOLOv8n-pose, detects candidates via open-vocabulary semantic detection, and asks a clarifying question whenever more than one candidate survives the pointing cone. It sits within a six-condition ablation that adds one interaction variable per condition and has passed through three perception stacks.

The comparison run show a clear result on the test frame. No model selected the correct referent when the prompt left the gesture unmentioned. Two of the three selected it as soon as the prompt directed them to the gesture, both landing within three pixels of ground truth. The pipeline that computes an explicit three-dimensional pointing ray selected the wrong object, and the one model that reproduced that same geometry in prose selected the same wrong object. On this frame, explicit geometry was therefore neither necessary nor sufficient for correct grounding, and what accounted for the outcome was the prompt condition and not the choice of model. The experiment covers one frame. Extending it to further frames is what remains.

## 1 Introduction

Instructions given to robots in shared physical spaces are frequently underspecified. A person does not say "pick up the mint green soap dispenser two metres to your left". They say "pick that up" and point. The linguistic content alone is insufficient, and the gesture alone is insufficient, so the referent has to be recovered by fusing the two with what the robot can see.

This problem is called deictic grounding. It is difficult for three reasons that compound each other. Pointing is imprecise, and a small change in arm angle can move the projected ray from one object to its neighbour. Scenes contain multiple plausible referents of the same category, so a correct object class does not determine a correct object instance. And the robot observes the gesture from its own viewpoint, which is not the viewpoint of the person producing it.

The objective of this work is to build a system that performs deictic grounding on a legged robot with an arm, that recognises when it cannot resolve the referent, and that asks a clarifying question instead of acting on a guess. A second objective is to establish whether frontier vision-language models, given the same frame and no geometric machinery, arrive at the same referent and the same caution.

This report treats the second objective as an experiment. One recorded frame is held fixed, the prompt given to each system is varied across three conditions, and the referent each system selects is measured in image coordinates against a known ground truth. Holding the frame constant is what makes the comparison interpretable, because every system then receives exactly the same pixels and the same instruction text apart from the one clause under manipulation.

The contributions of this study:

- A six-condition ablation for deictic grounding on a Boston Dynamics Spot, in which each condition adds exactly one interaction variable. The first three have been run on the robot. The three that involve robot motion are built and require further experimentation.
- A live pipeline that runs fully offline on the robot and that abstains from answering when more than one candidate survives the pointing cone.
- A replay benchmark, with a protocol fixed before any runs, that compares the pipeline against three frontier models on identical frames under three prompt conditions.
- A controlled single-frame comparison establishing that explicit geometry was neither necessary nor sufficient for correct deictic grounding.

## 2 Related Work

**Fusing language, gesture and vision.** LEGS-POMDP treats language, gesture and visual observation as separate signals that are fused probabilistically for open-world object search [1]. Its reported findings motivate the present design. Language alone is insufficient for deictic reference in robot instruction, combining language and gesture improves resolution accuracy, and the encoding chosen for the pointing gesture materially changes performance. Related work evaluates alternative pointing vectors and gives a probabilistic observation model for using such a vector in object search [2].

**Asking instead of guessing.** Ask-to-Clarify frames the passive pattern of receiving an instruction and executing it as the failure mode, and resolves ambiguity through multi-turn dialogue [3]. This is the behaviour the present system implements as an explicit abstention rule. Clark’s account of language use, common ground and joint action provides the wider framing, in which participants establish shared understanding and repair misunderstanding rather than simply issuing and executing commands [4].

**Benchmarks for pointing and gesture.** Several recent benchmarks bear directly on this task. EgoPoint-Bench evaluates egocentric pointing over more than eleven thousand samples and reports that state-of-the-art models ground pointing by leaning on visual proximity and object saliency, a failure its authors name referential hallucination [5]. EgoPointVQA and the accompanying HINT method interleave tokens derived from 3D hand keypoints with the video input, and HINT-14B reaches 68.1 percent averaged over six deictic reasoning tasks [6].

EcoG-Bench is the most directly relevant to this experimental study. It measures whether a model can bind an underspecified deictic utterance to the co-speech pointing stroke that disambiguates it, and it reports a wide gap between human performance at 96.9 percent and the best native video-audio model at 17.0 percent. It further reports that replacing the native interface with timestamped frame samples and externally verified speech recognition raises the same model from 17.0 to 42.9 percent, and concludes that the multimodal interface can bottleneck performance independently of model reasoning [7]. That conclusion is the large-sample analogue of what this report observes on one frame. EgoExoBench reports that multimodal language models fail at cross-perspective alignment even when they perform well on single-view tasks [8]. For data and annotation, NatSGD provides synchronised robot-egocentric and third-person views of the same interaction [9], NatSGLD adds speech, gesture, logic and demonstration for robot learning [10], and GESRes provides a gesture annotation schema that separates abstract deictic from regular deictic reference [11].

### 3 Methodology

#### 3.1 Study Design

The study is a six-condition ablation. Each condition adds exactly one variable to the preceding one, so that any change in behaviour is attributable to that variable.

Table 1: The six ablation conditions. Each condition is a subclass of a shared engine that flips a single feature flag. The last three conditions (4-6) that require the robot to move are implemented and have not yet been run.

| # | Condition     | Variable added                                                                                         | Status    |
|---|---------------|--------------------------------------------------------------------------------------------------------|-----------|
| 1 | Baseline      | Voice and typed input, pointing cone, sweep gestures. A question is asked but no answer turn is taken. | Runs live |
| 2 | Gaze prior    | Head-pose yaw re-weights the candidate set.                                                            | Runs live |
| 3 | Clarification | A spoken or typed answer turn closes the dialogue loop.                                                | Runs live |
| 4 | Repositioning | Consent-gated body twist and re-ground, without walking.                                               | Built     |
| 5 | Manipulation  | Consent-gated walk to the object and arm pick.                                                         | Built     |
| 6 | Exocentric    | An additional camera positioned at the human viewpoint.                                                | Built     |

Conditions 1, 2 and 3 have been run on the robot, and every recorded session drawn on in this report comes from them. Conditions 4, 5 and 6 are implemented against the same shared engine and have not been run. What separates them from the first three is robot motion. Repositioning and manipulation require Spot to twist and to walk, and the exocentric condition requires a second camera that has not yet been chosen.

#### 3.2 Platform and Perception Stacks

All experiments use a Boston Dynamics Spot with an arm, on SDK version 4.1.0. The gripper camera provides the colour image used for grounding, and five body fisheye cameras provide surrounding context. The perception stack has been rebuilt twice.

The division of labour between the detector and the VLM in v2 is that the detector owns coordinates and the language model owns semantics. The detector proposes boxes for whatever noun it is given, and the language model decides what those boxes depict.

#### 3.3 The Live Pipeline

The v2 stack runs entirely offline on the robot, which is a requirement because there is no network at the robot in the field. A command arrives either typed or spoken. Spoken commands are opened by the wake word “Spot” captured through a lapel microphone and transcribed locally. The pipeline then extracts a target noun and a scope from the sentence, captures a pointing or sweeping gesture, detects only that noun, and filters the detections through a cone constructed around the pointing ray. Figure 1 shows the pose skeleton, the projected ray and the resolved target on a live frame.

Table 2: Three perception stacks. Each replaced the previous one for a stated reason.

| Version | Perception                                                                                                                    | Rationale for the change                                                                                              |
|---------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| v1      | MediaPipe pose and gaze, with the language model reporting object boxes                                                       | The language model self-reported coordinates, which were imprecise, and no open vocabulary was available              |
| v2      | YOLOv8n-pose for the ray, YOLO-Worldv2 for open-vocabulary semantic object detection, local language model for semantics only | MediaPipe lost the person when they were small, distant or partly out of frame, which made the whole pipeline fragile |
| v3      | Raw frames to a frontier vision-language model, with no geometric priors                                                      | Isolates what the explicit geometric pipeline contributes                                                             |

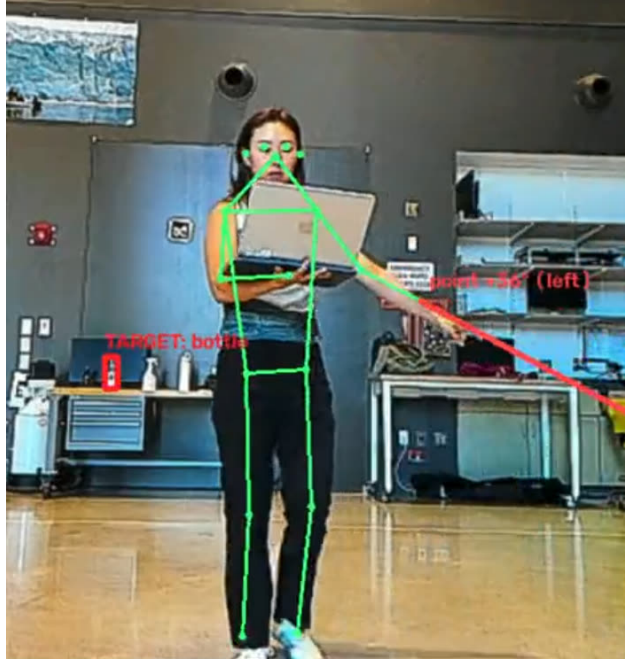


Figure 1: A live grounding frame. The pose skeleton and the projected pointing ray are drawn in green and red.

### 3.4 Abstention and Clarification

The system abstains rather than guessing. When exactly one candidate survives the cone, it commits. When more than one survives, it describes each candidate crop in natural language and asks which one is intended. If an answer arrives, it is matched back to a candidate. If no answer arrives, the system selects the candidate most aligned with the pointing ray, states aloud that it is guessing, and instructs the operator to repeat the command. This made failure visible instead of silent.

### 3.5 Operator Interface and Logging

Figure 2 shows the operator interface. It presents the body and gripper camera views, the pose overlay and ray, the candidate crops being sent to the language model, and the running system output. Motion is opt-in and consent-gated behind an emergency stop. A countdown indicator marks the window in which the microphone and the gesture capture are open, and a separate state marks the period during which the models are running, so that the operator can tell whether a long pause is computation or a failure.

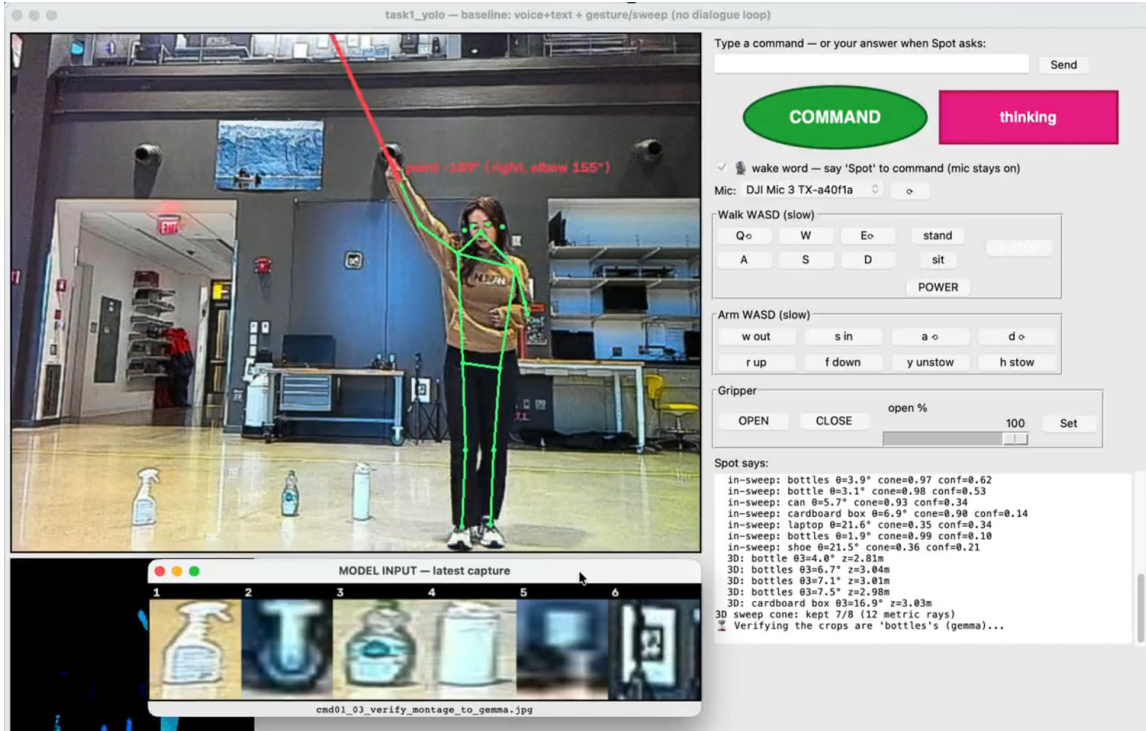


Figure 2: The operator interface during a sweep gesture. Camera views and overlays are on the left, the candidate crops sent to the language model are below, and the running system log and robotic controls are on the right.

Every session can be recorded. Per-camera video, captured grounding frames, and a session log are written for each run. The log records the selected referent with its fused confidence and angular offset from the ray, every clarifying question and answer, the planned action and the outcome.

### 3.6 Replay Benchmark

Live runs cannot be compared with one another, because the scene moves, the lighting shifts and the gesture is never repeated exactly. The benchmark therefore replays a single frozen frame through every system. Each frame and instruction is presented under three conditions. Three frontier models were evaluated against the v2 pipeline.

Table 3: The three prompt conditions.

| Condition        | What the model receives                                    |
|------------------|------------------------------------------------------------|
| Free choice      | Image and instruction, with no constraint on the answer    |
| Pointing only    | Instructed to return a coordinate                          |
| Pointing may ask | Told it may ask a clarifying question instead of answering |

### 3.7 Data Collection and Analysis

Ground truth is annotated by the person who produced the gesture, recorded before the system responds, so that the intended referent cannot be contaminated by the system output. Each episode stores the frame, the labelled object boxes, the intended target, and the live system’s own selection and reasoning trace.

The planned measures are referent accuracy and confidence at the decision point.

## 4 Results and Discussion

### 4.1 Experimental Design

The experiment holds one recorded frame fixed and varies the prompt. The frame shows a person pointing at three visually similar bottles on the floor, paired with the spoken instruction “pick that up”. The ground truth referent is the bottle furthest from the person, on the left of the frame, at a horizontal image position of approximately 77 pixels. Three near-identical objects of the same category were chosen deliberately. Object class cannot separate them, so the gesture is the only signal in the scene that can.

Table 4: Structure of the single-frame experiment.

| Role        | Variable                                                                                                                                                                                                                                                                                  |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Held fixed  | The image, the scene, the instruction “pick that up”, and the ground truth referent                                                                                                                                                                                                       |
| Manipulated | The prompt condition, at three levels. <i>Free choice</i> places no constraint on the answer. <i>Pointing only</i> directs the system to the pointing gesture and requests a coordinate. <i>Pointing may ask</i> adds permission to return a clarifying question in place of a coordinate |
| Measured    | The horizontal image coordinate of the selected referent, whether a clarifying question was returned instead, and end-to-end latency                                                                                                                                                      |

Four systems answer. Three are frontier vision-language models that receive raw pixels and carry no geometric machinery, and each answers under all three prompt conditions. The fourth is the robot pipeline of Section 3, which computes an explicit three-dimensional pointing ray. It runs under its own single configuration, because its prompt is fixed by its implementation.

### 4.2 Results

Figure 3 overlays every answer on the frame.

Table 5: Referent selection by model and condition on the example frame. The correct referent is the leftmost bottle, at a horizontal image position of approximately 77 pixels. “Asked” denotes a clarifying question returned in place of a coordinate.

| Condition        | GPT-5              | Molmo2-ER          | Gemini Robotics-ER |
|------------------|--------------------|--------------------|--------------------|
| Free choice      | 230 px (incorrect) | 180 px (incorrect) | Asked              |
| Pointing only    | 77 px (correct)    | 80 px (correct)    | 179 px (incorrect) |
| Pointing may ask | 77 px (correct)    | 77 px (correct)    | Asked              |

**Finding 1. The prompt changed which object was selected.** In the free condition, no system selected the correct referent. GPT-5 chose a bottle at 230 px and Molmo2-ER chose one at 180 px, against a ground truth of 77 px. Adding a clause that directed the system to the pointing gesture moved both to the correct bottle, at 77 px and 80 px. Adding permission to ask a question moved both to 77 px. Other than the instruction, nothing about the image, the scene or the models changed across these three conditions.

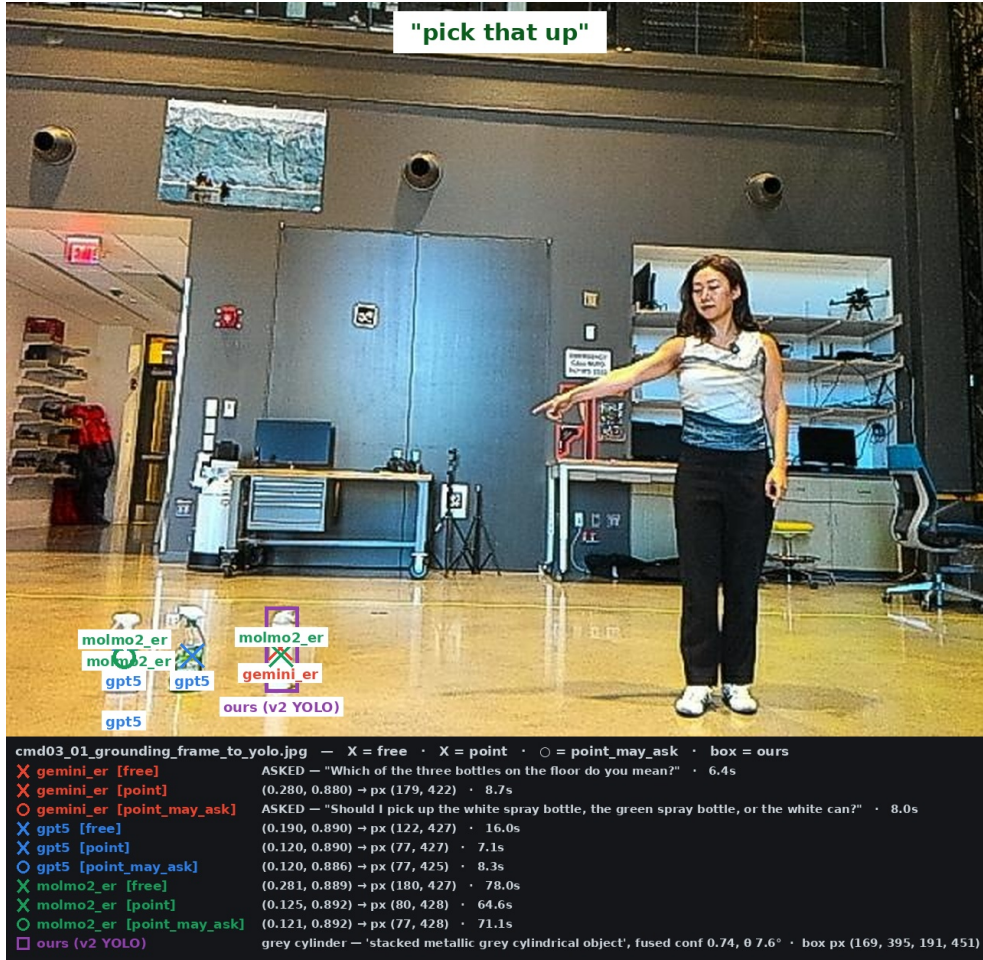


Figure 3: Every system’s answer to “pick that up”, overlaid on one frame. Marker shape encodes the prompt condition and the box is the geometric pipeline’s selection.

**Finding 2. Explicit geometry was not sufficient.** The robot pipeline extracts a three-dimensional pointing ray from pose keypoints, intersects it with a candidate set and reports a fused confidence. On this frame it selected the wrong bottle, at a fused confidence of 0.74 and an angular offset of 7.6 degrees from the ray. Gemini Robotics-ER selected that same wrong bottle in the pointing-only condition, and its stored reasoning trace describes extracting shoulder, elbow and wrist positions, computing a slope, extending the vector and concluding a referent. One system performed that geometry numerically and the other performed it in prose, and both arrived at the same incorrect object.

**Finding 3. Explicit geometry was not necessary.** GPT-5 and Molmo2-ER carry no pose estimator, no camera model and no ray intersection. Both selected the correct referent under both pointing conditions, and both landed within three pixels of ground truth. Whatever these models do when they are directed to a gesture recovered the referent on a frame where an explicit ray did not.

**Finding 4. One model declined to answer when nothing invited it to.** Gemini Robotics-ER returned a clarifying question in the free condition, where the prompt offered no such option, and again in the condition that permitted it. This is the behaviour the robot pipeline produces through an explicit abstention rule when more than one candidate survives the cone, reached here with no rule and no geometry.

## 5 Limitations

Due to time limitations, no condition involving robot motion has been evaluated. Repositioning, walking and manipulation are implemented and have not been run, so every observation in this report concerns a stationary robot resolving a referent from one viewpoint. Whether a second viewpoint reduces the number of surviving candidates, and whether it therefore lowers the ask rate, is an open question.

## 6 Future Work

The study will continue with annotating the remaining recorded grounding frames into scored episodes and running the same three prompt conditions across all of them, which turns the counterexamples reported here into rates. The second task is implementing the ask-rate measure in the scoring code.

In addition, the last three of the six ablation conditions will have to be tested. Each one is gated on the same thing, which is a robot session with motion enabled. Each one also tests a question the stationary system cannot reach.

**Condition 4, repositioning.** A consent-gated body twist gives a second viewpoint on the same scene, and the pipeline re-grounds from that view. The question is whether a second view separates candidates that a single view leaves tied. If it does, the correct response to ambiguity is sometimes to move and look again instead of asking.

**Condition 5, manipulation.** Walking to the resolved object and picking it up closes the path from reference to action. It also introduces the first real cost of being wrong. Up to this point an incorrect referent produces an incorrect label in a log. Once the arm moves, an incorrect referent produces an incorrect grasp.

**Condition 6, exocentric camera.** A second camera placed near the human viewpoint tests whether the cross-perspective gap reported by EgoExoBench [8] shows up in this task. The robot currently observes a gesture from a viewpoint that is not the one producing it, and this condition is the only one that removes that asymmetry.

## 7 Conclusion

This project set out to build a robot that can resolve an underspecified spoken instruction against a pointing gesture, and that knows when it cannot. That system now exists and runs. It works entirely offline on a Boston Dynamics Spot, it recovers the pointing ray from the robot’s own camera, it detects candidate objects with an open vocabulary, and it asks a clarifying question whenever more than one candidate survives the pointing cone. Around it sits a six-condition ablation in which each condition isolates a single interaction variable, and a replay benchmark that puts the pipeline and three frontier vision-language models in front of identical frames under identical prompts.

That benchmark was then used to run an experiment. One frame was held fixed, the prompt was varied across three conditions, and the referent each system selected was measured in image coordinates against a known ground truth. The result on that frame is unambiguous. No system selected the correct bottle while the prompt left the gesture unmentioned. Two of the three selected it as soon as the prompt named the gesture, both landing within three pixels of ground truth. The one system built around an explicit three-dimensional pointing ray selected the wrong bottle, at a high fused confidence of 0.74 and a small angular offset of 7.6 degrees, and the one model that reproduced that same geometry in prose selected the same wrong bottle.

Explicit geometry was therefore neither necessary nor sufficient for correct deictic grounding on this frame. Both are counterexample claims, and one frame settles a counterexample. The finding matters for how these systems are built, because the prevailing architecture puts a geometric module in front of a language model on the assumption that recovering the geometry is the hard part. Here the geometry was recovered, reported with confidence, and wrong, while two models holding no geometry at all were right.

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